Large-scale cortical dynamics across anesthesia and sleep in macaque ECoG

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Abstract

General anesthesia and sleep alter large-scale brain dynamics, but it remains unclear which ECoG changes are common to these states. This study analyzed publicly available NeuroTycho recordings from two macaques. Eight paired comparisons contrasted awake epochs with epochs recorded during ketamine, medetomidine, propofol anesthesia, or natural sleep. Large-scale ECoG dynamics were assessed using phase synchrony, metastability, phase-lagged connectivity, and dynamic network measures. The most consistent change was an increase in broadband wPLI, indicating stronger phase-lagged functional coupling during anesthesia and sleep. Delta-band wPLI also increased, while theta-band Kuramoto synchrony and delta-band metastability decreased. Several effects were observed before correction, but none remained significant after false discovery rate correction. The results suggest that anesthesia and sleep share some changes in large-scale ECoG dynamics. Broadband phase-lagged coupling showed the most consistent pattern, while spatial and dynamic changes differed across conditions.

Keywords: consciousness; anesthesia; sleep; electrocorticography; macaque; weighted phase lag index; phase synchronization

1 Introduction

Wakefulness depends not only on neural activity but also on how this activity is organized across cortical regions. In the awake brain, large-scale cortical activity is structured but flexible. Patterns of interaction between regions change over time. During unconscious states, such as general anesthesia and sleep, this coordination is altered. Interactions across cortical regions may become less efficient, less diverse, or organized in different ways [Alkire et al., 2008, Brown et al., 2010].

Previous studies have linked anesthesia, sleep, and reduced consciousness to changes in functional connectivity, temporal complexity, and transient network dynamics [Barttfeld et al., 2015, Chen et al., 2022, Ensel et al., 2024]. Together, these findings suggest that states of consciousness are reflected in large-scale brain dynamics. However, anesthesia and sleep differ in their different physiological mechanisms. It therefore remains unclear which changes in brain dynamics are shared across these states and which are specific to each state.

Macaque electrocorticography (ECoG) is well suited to address this question because it provides high temporal resolution and broad cortical coverage. The NeuroTycho dataset contains recordings from two macaques during wakefulness, natural sleep, and anesthesia, which are induced by ketamine, medetomidine, and propofol [Yanagawa et al., 2013]. This study used these public recordings to compare wakefulness with anesthesia and sleep. The analysis focused on both shared and condition-specific changes in large-scale ECoG activity.

The dataset included eight paired comparisons. Each comparison contrasted recordings from the same monkey during wakefulness and either anesthesia or sleep. Because the number of comparisons was limited, the study focused on patterns that were consistent within the dataset rather than making population-level inferences. The goal was to describe large-scale cortical dynamics and identify the ECoG features that were most consistent across comparisons.

2 Materials and methods

2.1 Dataset

This study used publicly available macaque ECoG recordings from NeuroTycho [Yanagawa et al., 2013]. The data were collected from two macaques, George and Chibi, during wakefulness, drug-induced anesthesia, and natural sleep. ECoG signals were recorded from 128 electrodes at a sampling rate of 1000 Hz.

The analysis included four non-awake conditions: ketamine anesthesia, medetomidine anesthesia, propofol anesthesia, and natural sleep. Here, the term ”unconsciousness” collectively refers to the periods of anesthesia and sleep involved in the analysis. For each monkey and condition, the analysis retained one awake epoch and one unconscious epoch. This resulted in eight paired comparisons. In the anesthesia data, the awake and unconscious epochs were labeled *AwakeEyesOpened* and *Anesthetized*, respectively. In the sleep data, they were labeled *AwakeEyesOpened* and *Sleeping*, respectively.

2.2 Epoch and channel selection

For each monkey-condition pair, epochs were selected using the behavioral labels provided with the original dataset. Each

recording session was treated as a separate block with its own time axis. Epochs from different sessions were not concatenated. For each state, the analysis retained the longest continuous segment with the relevant label. The labels were *AwakeEyesOpened* for wakefulness, *Anesthetized* for anesthesia, and *Sleeping* for sleep. This yielded one awake epoch and one unconscious epoch for each paired comparison. Epoch duration varied across comparisons because the labeled segments differed in length. The source and duration of each epoch are reported in Supplementary Table S1.

The main analysis used a fixed subset of 64 channels from the original 128-channel ECoG array. These channels were selected at regular intervals across the implanted array before bad channel detection. The same subset was used for all monkey-condition pairs. This subset provided broad spatial sampling across the implanted cortical area while reducing the computational cost of subsequent analyses. Bad channels were then identified separately for each monkey-condition pair. For each monkey-condition pair, channels were excluded if they had extreme variance or if their time series showed a weak correlation with the mean signal across channels. All main analyses used the remaining good channels. As a robustness check, the study repeated the analysis using all 128 ECoG channels. The fixed 64-channel subset and electrode layouts are shown in Supplementary Figure S1.

2.3 Preprocessing and phase extraction

Raw ECoG signals were downsampled from 1000 Hz to 250 Hz before analysis. The highest frequency examined was 50 Hz, which was below the Nyquist frequency of 125 Hz. The signals were then rereferenced to the common average of the retained good channels for each monkey-condition pair [Ludwig et al., 2009].

Instantaneous phase was estimated for five frequency bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and broadband activity (1-50 Hz). For each frequency band, the rereferenced signals were band-pass filtered using zero-phase Butterworth filtering. The Hilbert transform was then applied to obtain the analytic signal. The instantaneous phase of each channel was extracted from this signal and used to calculate the phase-based metrics [Cohen, 2014, Widmann et al., 2015].

2.4 Phase-based metrics

Kuramoto synchrony and metastability. Large-scale phase coordination was measured using the Kuramoto order parameter [Kuramoto, 1984, Cumin and Unsworth, 2007]. This parameter describes the alignment of instantaneous phases across the retained ECoG channels. For each frequency band and state, it was calculated as

$$R(t) = \left| \frac{1}{N} \sum_{j=1}^{N} e^{i\phi_j(t)} \right|,$$

where N is the number of retained good channels and $\phi_j(t)$ is the instantaneous phase of channel j at time t. The value of $R(t)$ approaches 1 when phases are strongly aligned across channels. It approaches 0 when phases are more widely dispersed. Kuramoto synchrony was defined as the temporal mean of $R(t)$ and represented average global phase alignment. Kuramoto metastability was defined as the temporal variance of $R(t)$ and represented changes in global phase coordination over time.

Phase-lagged functional connectivity. Phase-lagged functional connectivity was measured using the weighted phase lag index (wPLI) [Vinck et al., 2011, Malekmohammadi et al., 2019]. wPLI measures the consistency of non-zero-lag phase coupling between two signals. It is less sensitive to zero-lag correlations that may arise from common sources or volume conduction. In this study, functional connectivity refers to statistical coupling between ECoG signals rather than anatomical connectivity.

For channels i and j, wPLI was calculated as

$$\mathrm{wPLI}_{ij} = \frac{\left|\mathbb{E}\left[\mathrm{Im}(S_{ij})\right]\right|}{\mathbb{E}\left[\left|\mathrm{Im}(S_{ij})\right|\right]},$$

where S_{ij} is the cross-spectrum between the two channels and $\mathrm{Im}(\cdot)$ denotes its imaginary component.

For each monkey-condition pair, state, and frequency band, wPLI was calculated for every unique pair of retained good channels. Global wPLI was defined as the mean across all channel pairs. For the topographic analysis, each channel was assigned a node-strength value. This value was defined as the mean wPLI between that channel and all other retained good channels. For each channel, the state-related topographic change was calculated as broadband wPLI node strength in the unconscious state minus that in the awake state.

2.5 Additional dynamic network metrics

Functional connectivity dynamics (FCD) variance was used to summarize how phase-based connectivity patterns varied across time [Hansen et al., 2015, Barttfeld et al., 2015]. For each epoch, phase-based connectivity matrices were computed from instantaneous phase values in 5-s sliding windows with a 3-s step. The upper-triangular entries of each windowed matrix were vectorized, and pairwise similarities between windowed connectivity patterns were used to form the FCD matrix. FCD variance was defined as the variance of the off-diagonal entries of this FCD matrix.

Coalition entropy was used to summarize the diversity of transient phase-coordination patterns [Schartner et al., 2015]. For each 2-s sliding window with a 1-s step, a phase-coherence matrix was computed and its leading eigenvector was extracted. The sign pattern of this eigenvector defined a transient coalition state. Coalition entropy was computed as the normalized entropy of the distribution of coalition states across windows.

2.6 Statistical analysis and robustness checks

Statistical analyses were based on eight paired awake-to-unconscious comparisons, covering two monkeys and four unconscious conditions. For each metric and frequency band, every monkey-condition pair provided one awake value and one unconscious value. State effects were defined as the unconscious-minus-awake difference.

Because only eight paired comparisons were available, state differences were tested using exact paired sign-flip tests. These tests do not require a normality assumption [Maris and Oostenveld, 2007]. Effect sizes were reported as paired Cohen's d and were calculated from the unconscious-minus-awake differences [Cohen, 2013]. Bootstrap 95% confidence intervals were used to describe uncertainty in the effect-size estimates.

False discovery rate correction was applied across all 17 metric-band tests to account for multiple comparisons [Benjamini and Hochberg, 1995]. Bayes factors for paired samples were also calculated. They measured the evidence for a state difference relative to the null hypothesis of no difference [Rouder et al., 2009]. In the Results, d denotes paired Cohen's d, p denotes the uncorrected p value from the exact paired sign-flip test, and BF_{10} denotes the Bayes factor in favor of a state difference.

Two robustness checks were conducted. First, epoch duration varied across monkey-condition pairs. To control for this variation, the analysis was repeated using one 120-s segment from each retained awake and unconscious epoch. The same preprocessing steps, good-channel masks, and metric definitions were used as in the main 64-channel analysis. Second, the analysis was repeated using all 128 ECoG channels instead of the fixed 64-channel subset.

3 Results

3.1 Core changes in cortical dynamics across state transitions

Figure 1 summarizes the study design, metrics, and main effect-size results. The dataset provided eight paired awake-to-unconscious comparisons across two macaques. The unconscious conditions included ketamine, medetomidine, and propofol anesthesia, as well as natural sleep. The 17 metric-band tests covered three aspects of ECoG dynamics: phase coordination, phase-lagged coupling, and dynamic organization.

Broadband wPLI showed the largest difference between awake and unconscious epochs. In the main 64-channel analysis, broadband wPLI was higher during unconscious epochs (paired Cohen's $d = 1.547$, exact paired sign-flip $p = 0.0156$, $BF_{10} = 21.284$). Since global wPLI was calculated as the mean across all retained good-channel pairs, the increase therefore indicates stronger network-level phase-lagged coupling during unconscious epochs.

Three additional state differences had uncorrected p values below 0.05, but their effect sizes were smaller than that of broadband wPLI. Delta-band wPLI increased during unconscious epochs ($d = 1.160$, $p = 0.0156$, $BF_{10} = 7.432$). In contrast, theta-band Kuramoto synchrony decreased ($d = -1.092$, $p = 0.0312$, $BF_{10} = 6.107$). Delta-band Kuramoto metastability also decreased ($d = -0.962$, $p = 0.0312$, $BF_{10} = 4.156$). Together, these effects were characterized by stronger phase-lagged coupling, lower theta-band global phase alignment, and less temporal variation in delta-band global coordination during unconscious epochs.

However, none of the 17 metric-band tests remained significant after FDR correction. The smallest FDR-adjusted p value was 0.1328. Complete numerical results are provided in Supplementary Table S2. These findings should therefore be treated as exploratory. Among the tested metrics, broadband wPLI showed the largest state-related effect.

3.2 Broadband coupling and spatial redistribution

Figure 2 shows the frequency and spatial patterns of change in phase-lagged coupling. In Panel A, broadband wPLI increased in all four unconscious conditions. Changes in the narrower frequency bands were smaller and varied more across conditions. Delta-band wPLI showed modest increases. Theta-band and alpha-band wPLI showed little difference between states, while beta-band changes varied across conditions. The main frequency pattern was therefore a broadband increase rather than a similar increase across all frequency bands.

Panel B shows the spatial distribution of changes in broadband wPLI node strength. Node strength was defined as the mean wPLI between each electrode and all other retained electrodes. For each electrode, the plotted value was calculated as node strength during the unconscious epoch minus that during the corresponding awake epoch. These changes were not uniform across the implanted arrays. Ketamine, medetomidine, and propofol generally produced positive changes in parts of the arrays. However, the location and size of these changes differed across monkeys and conditions. Medetomidine produced the most widespread positive changes, whereas sleep produced weaker and more variable patterns.

Together, these results suggest that unconscious states were associated with stronger broadband phase-lagged coupling at the network level. The spatial pattern of this change varied across the recorded cortical areas. The maps summarize changes at the electrode level and should not be interpreted as precise anatomical localization within a standard brain atlas.

3.3 Shared and condition-specific patterns across unconscious conditions

Broadband wPLI was the most consistent shared change across ketamine, medetomidine, propofol, and sleep. In the main 64-channel analysis, it increased in seven of the eight paired awake-to-unconscious comparisons (Fig. 3B). Thus, most comparisons showed stronger broadband phase-lagged coupling during the unconscious state. Changes in individual pairs and condition-level summaries are provided in Supplementary

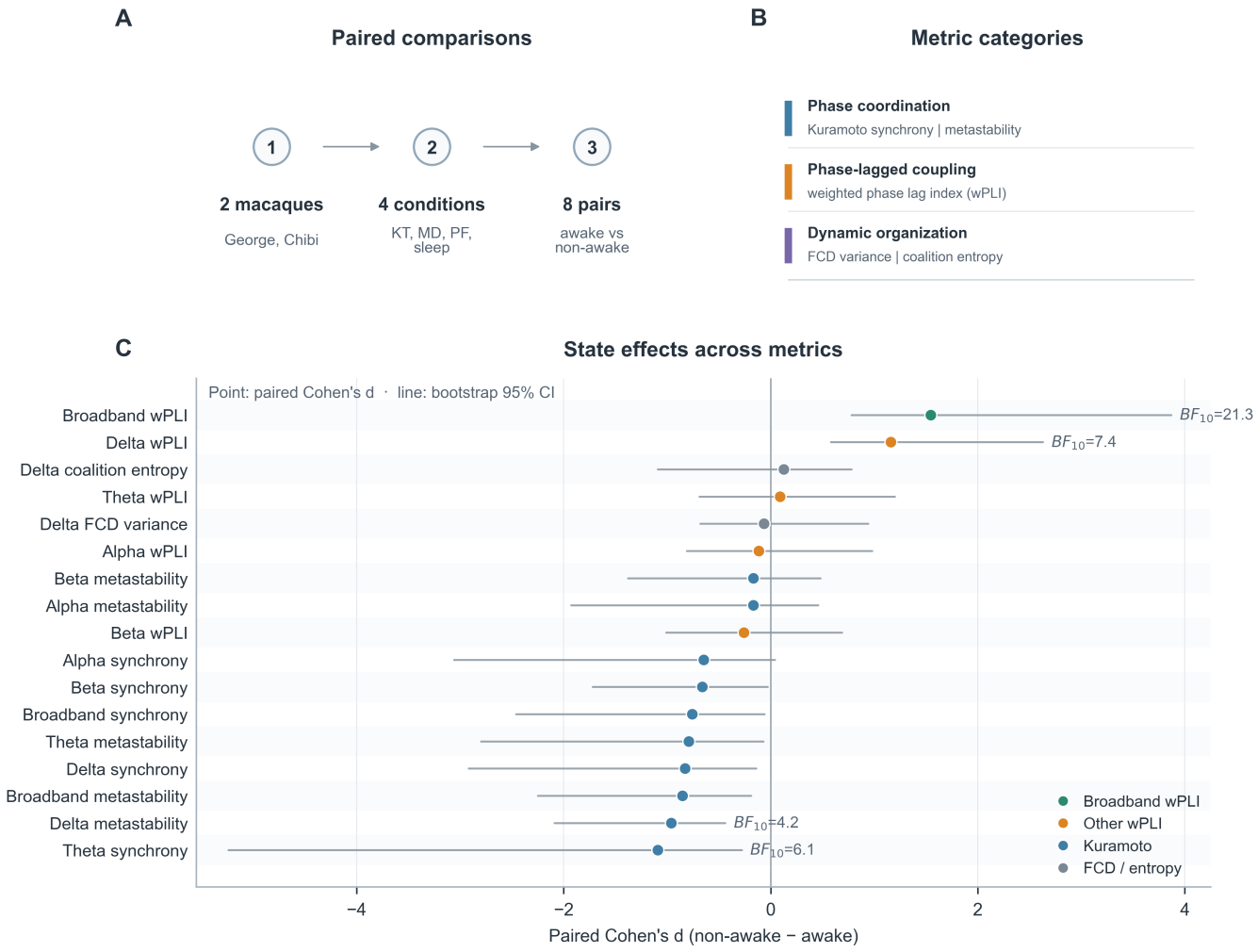


Figure 1: Study overview and effect-size summary. **A**, Eight paired awake-to-unconscious comparisons from two macaques and four unconscious conditions. **B**, Metrics grouped into phase coordination, phase-lagged coupling, and dynamic organization. **C**, Effect sizes and bootstrap 95% confidence intervals for all 17 metric-band tests. Effects were calculated as unconscious minus awake, so positive values indicate higher values in the unconscious state. Broadband wPLI had the largest positive effect size, although no test remained significant after FDR correction.

Figure S2 and Supplementary Table S3.

However, the spatial pattern of broadband wPLI changes differed across conditions and monkeys (Fig. 2B). Medetomidine produced the most widespread increases in node strength. Sleep produced weaker and more spatially mixed changes. Ketamine and propofol also increased node strength in several parts of the implanted arrays, but these changes were more spatially uneven. These results suggest that broadband wPLI generally increased across conditions, although the spatial distribution of this increase differed between conditions.

The Kuramoto measures also showed common directional changes. Theta-band synchrony and delta-band metastability each decreased in seven of the eight paired comparisons (Fig. 3B). However, these effects were less stable in the robustness analyses and should be interpreted with caution. Overall, increased broadband phase-lagged coupling was the most consistent shared change, whereas the decreases in theta-band synchrony and delta-band metastability were less robust.

3.4 Robustness of state-related changes in broadband wPLI

Two robustness analyses tested the effects of epoch duration and channel selection. The first used one 120-s segment from each awake and unconscious epoch. The second used all 128 ECoG channels instead of the 64-channel subset. Broadband wPLI remained higher during unconscious epochs in all analysis settings (Fig. 3A). In the 64-channel analysis of 120-s segments, the effect size was $d = 1.605$ (uncorrected $p = 0.0156$). The effect was also present in the 128-channel analysis. This was the case for both the full epochs ($d = 1.779$, uncorrected $p = 0.0078$, $BF_{10} = 37.138$) and the 120-s segments ($d = 1.656$, uncorrected $p = 0.0078$, $BF_{10} = 27.845$). Complete results for these robustness analyses are provided in Supplementary Tables S4 and S5.

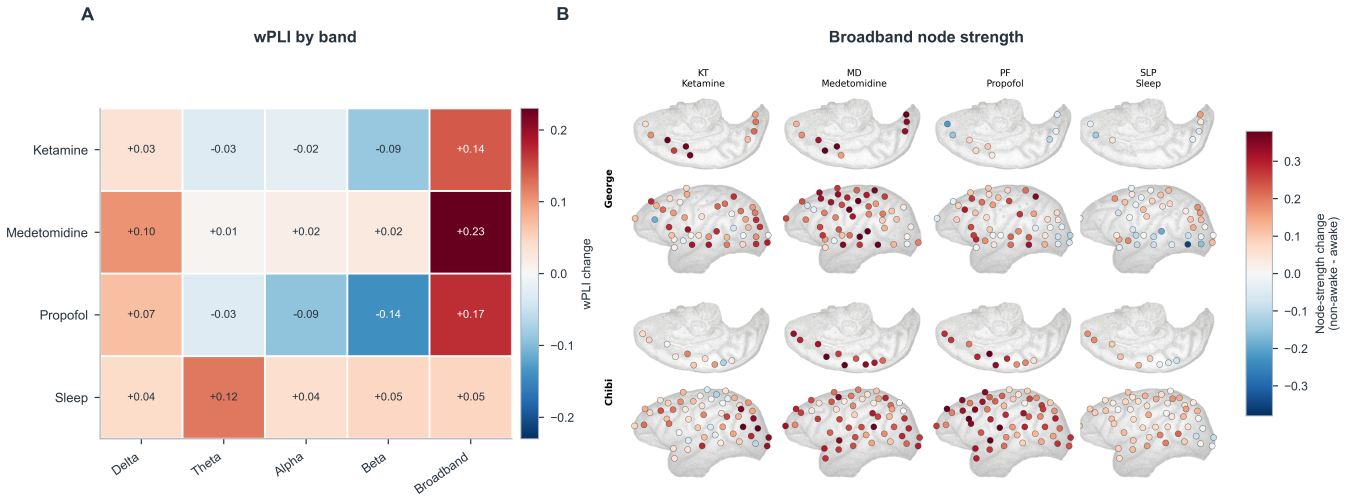


Figure 2: Frequency and spatial patterns of wPLI changes. **A**, Mean changes in wPLI across frequency bands. Changes were calculated as unconscious minus awake and averaged across the two macaques. **B**, Changes in broadband wPLI node strength shown on the two-dimensional NeuroTycho ECoG maps. Warm colors indicate higher node strength during the unconscious state, whereas cool colors indicate lower node strength.

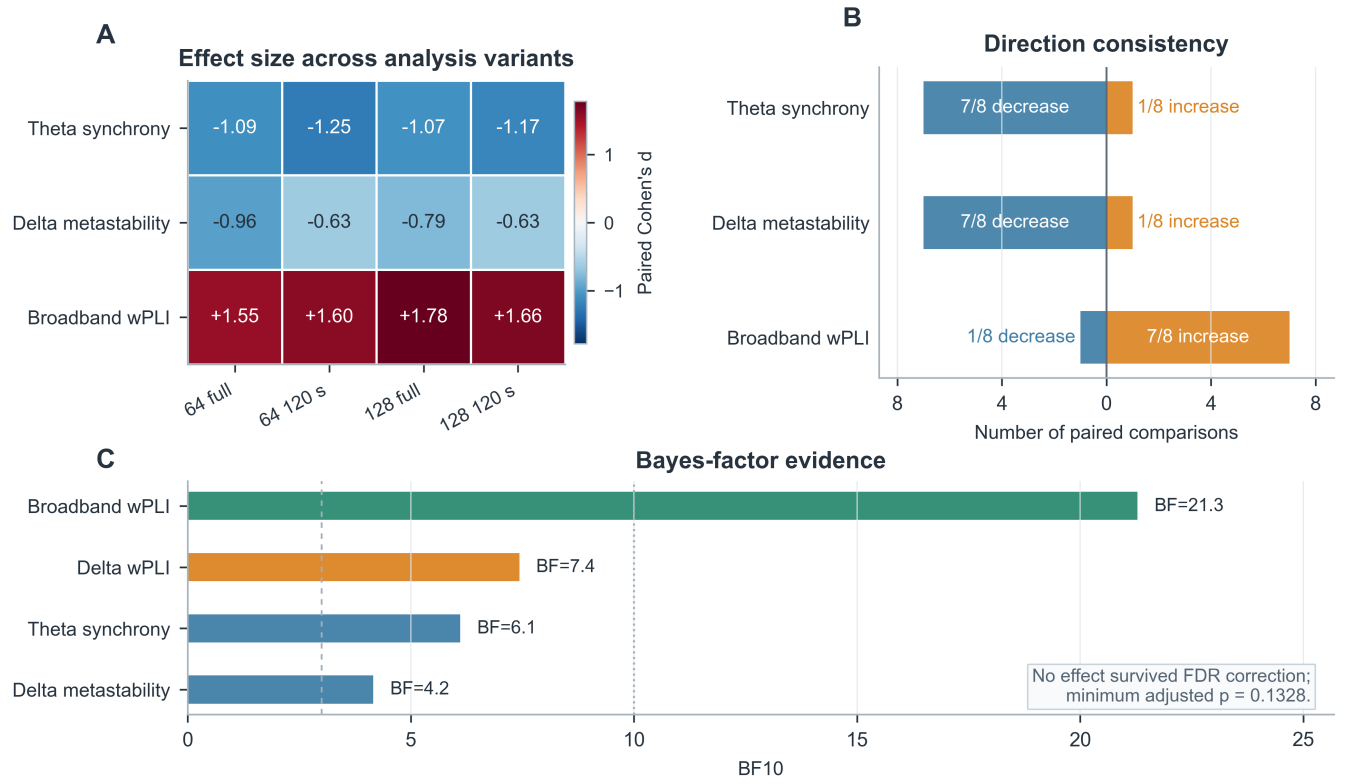


Figure 3: Robustness and evidence summary. **A**, Paired Cohen's d values for the three main effects across four analysis settings: 64 or 128 channels and full-length or 120-s epochs. **B**, Direction of change across the eight paired awake-to-unconscious comparisons in the main analysis. **C**, Bayes factors for the main 64-channel full-epoch analysis. No test remained significant after FDR correction. The smallest adjusted p value was 0.1328.

In contrast to broadband wPLI, the Kuramoto-based effects were less stable across the robustness analyses. For example, delta-band metastability remained lower in the 120-s analysis, but the effect was smaller ($d = -0.627$, uncorrected $p = 0.1875$).

Bayes factors showed the strongest evidence for a state dif-

ference in broadband wPLI (BF_{10} = 21.284). This was followed by delta-band wPLI, theta-band synchrony, and delta-band metastability (Fig. 3C). However, none of the 17 metric-band tests remained significant after FDR correction. Broadband wPLI was therefore the most stable pattern across the robustness analyses, but the finding should still be treated as exploratory.

4 Discussion

4.1 Principal finding

The main finding was a consistent increase in broadband phase-lagged coupling from wakefulness to anesthesia or sleep. Broadband wPLI increased in most paired monkey-condition comparisons. It also remained higher in analyses using 120-s segments and all 128 ECoG channels. This finding may appear unexpected because unconscious states are often linked to weaker large-scale cortical interactions [Alkire et al., 2008, Brown et al., 2010]. The result suggests that some forms of functional coupling may increase during anesthesia and sleep. However, wPLI measures consistent phase-lagged relationships between signals. A higher wPLI therefore indicates stronger functional coupling, but does not imply stronger anatomical connectivity between cortical regions.

At the same time, this finding should be interpreted cautiously. Broadband wPLI showed the clearest difference between awake and unconscious epochs, but none of the 17 metric-band tests remained significant after FDR correction. The broadband wPLI result should therefore be treated as the main exploratory finding in this dataset rather than as a confirmed effect after correction for multiple comparisons.

4.2 Shared and condition-specific organization across anesthesia and sleep

Ketamine, medetomidine, propofol, and sleep were each compared with wakefulness. However, the large-scale ECoG changes were not identical across conditions. Higher broadband wPLI was the clearest shared pattern. This suggests that increased phase-lagged coupling was common to most awake-to-unconscious comparisons in this dataset. In contrast, the spatial patterns and Kuramoto-based measures varied more across conditions.

The topographic maps showed clear differences between conditions. Medetomidine produced the most widespread increases in broadband node strength. Sleep produced weaker changes, with both increases and decreases across electrodes. Ketamine and propofol also increased node strength in parts of the implanted arrays, but their spatial patterns were less uniform. Thus, these maps provide electrode-level spatial summaries rather than precise anatomical localization. These spatial differences show variation across conditions but do not identify the physiological mechanisms underlying that variation.

Overall, increased broadband wPLI was the clearest shared pattern, whereas its spatial distribution and the Kuramoto-based changes were more condition-specific.

4.3 Interpretation of phase coordination and dynamic network summaries

The Kuramoto measures provided a different view of cortical dynamics from broadband wPLI. Theta-band synchrony was lower during unconscious epochs, indicating reduced large-scale phase alignment in this band. Delta-band metastability was also lower in the main analysis. This indicates less temporal variation in global phase coordination. However, the effect became smaller when 120-s segments were used. Delta-band metastability therefore appeared more sensitive to epoch duration than broadband wPLI.

These metrics describe different aspects of cortical dynamics. wPLI measures phase-lagged coupling between channel pairs. Kuramoto synchrony measures average phase alignment across channels, whereas Kuramoto metastability measures how this alignment varies over time. FCD variance and coalition entropy showed weaker differences between awake and unconscious epochs. Thus, not all measures of dynamic network organization distinguished the two states equally. Among the tested metrics, broadband wPLI showed the clearest and most stable pattern in this dataset.

4.4 Limitations and future directions

The findings should be considered within the size and structure of the dataset. The analysis included eight paired comparisons from two macaques across ketamine, medetomidine, propofol, and natural sleep. The paired design allowed direct comparisons between awake and unconscious states within each animal. Larger samples will help determine whether the same patterns occur in other animals.

Future studies should include a larger number of animals and involve multiple recording sessions. Such data will help determine whether the findings are reproducible across different animals, recording periods, and experimental settings. Furthermore, more detailed behavioral annotation could help reveal whether the changes in phase-lagged coupling during the loss and recovery of the waking state are gradual or abrupt.

5 Conclusion

In this macaque ECoG dataset, increased broadband phase-lagged functional connectivity was the most consistent difference between awake and unconscious epochs. Theta-band synchrony and delta-band metastability also decreased, but these changes were less stable across analyses. The broadband increase was shared across anesthesia and sleep, while its spatial distribution and other dynamic patterns varied by condition. These exploratory findings provide a basis for further studies of phase-lagged coupling and phase coordination during wakefulness, anesthesia, and sleep.

Data availability

The macaque ECoG recordings analyzed in this study are publicly available from NeuroTycho (http://neurotycho.org/). The specific datasets, sessions, and retained epochs used in the analysis are listed in Supplementary Table S1.

Ethics statement

This study used publicly available macaque ECoG recordings from NeuroTycho and did not involve any new animal experiments. Animal care and experimental procedures for the original recordings were described in the source dataset publication.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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